

WIRELESS EQUIPMENT CONTROL USING AT89C51

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Here is a microcontroller-based wireless equipment controller that can switch on or switch off up to four devices at a desired time interval set by the user in the transmitter. The devices can be controlled remotely from a distance of up to 30 metres from the transmitter. In the transmitter, an LCD module is used to show the device numbers and preset control time for the devices (00 to 99 seconds). Concepts of wireless RF communication and automation with AT89C51 microcontroller are used here.

The system is small, simple, cost-

effective and good for wireless control of home appliances or industrial instrumentation.

Block diagram

The system comprises a transmitter and a receiver as described below.

Transmitter section. Fig. 1 shows the block diagram of the transmitter section.

Four pushbutton switches (S1 through S4) are used as inputs to select the devices and set the time-out in the transmitter section. These are designated as up, down, enter and run keys, respectively. The time-out data is transferred over the RF wireless link to the receiver section.

The 8-bit AT89C51 microcontroller

is the main controlling part of the transmitter section. It is connected to the LCD module, input switches and encoder IC (HT12E). The device control program is stored in the memory of the microcontroller to control the devices as per the time-out settings done through input switches S1 through S4.

A two-line, 16-character LCD module shows the status of the main program that is running inside the microcontroller.

The HT12E is an 18-pin DIP package encoder IC that encodes 4-bit data and sends it to TRX-434 RF transmitter module.

The TRX-434 RF transmitter module uses a digital modulation

technique called amplitude-shift keying (ASK) or on-off keying. In this technique, whenever logic '1' is to be sent, it is modulated with carrier signal



RF Module Specifications

Parameter	Value
Frequency of operation	434 MHz
Modulation	ASK
Range	9.14 metres
Power Supply	5V (RX) 3V to 12V (TRX)

PARTS LIST

Semiconductors:

- IC1 - HT12E encoder
- IC2 - AT89C51 microcontroller
- IC3 - 7805 5V regulator
- IC4 - HT12D/SM5172 decoder
- IC5 - CD4519 multiplexer
- IC6 - ULN2003 relay driver
- TRX-434 - 434MHz RF transmitter module
- RX-434 - 434MHz RF receiver module
- LED1-LED8 - 5mm light-emitting diode
- D1-D10 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R17, R20-R24 - 1-kilo-ohm
- R3 - 680-kilo-ohm
- R4-R11 (RNW1) - 10-kilo-ohm network resistor
- R12-R15 - 4.7-kilo-ohm
- R16 - 1.2-kilo-ohm
- R18, R19 - 47-kilo-ohm
- VR1 - 47-kilo-ohm preset

Capacitors:

- C1 - 10 μ F, 16V electrolytic
- C2, C3 - 27pF ceramic
- C4 - 1000 μ F, 25V electrolytic
- C5 - 0.1 μ F ceramic
- C6 - 470nF ceramic

Miscellaneous:

- LCD1 - 16 $\times$ 2 line LCD display
- S1-S5 - Tactile switches
- X₁ - 11.0592MHz crystal
- J5 - Jumper switch
- Antenna - 4 $\times$ 2-pin berg strip male and female connectors

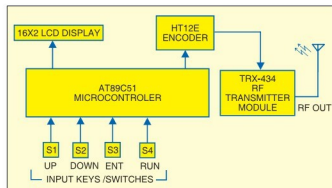


Fig. 1: Block diagram of transmitter section for wireless equipment control

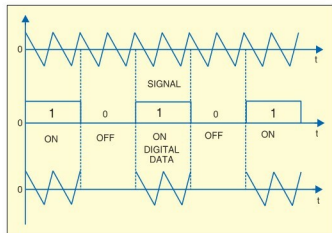


Fig. 2: ASK concept for the RF transmitter module